

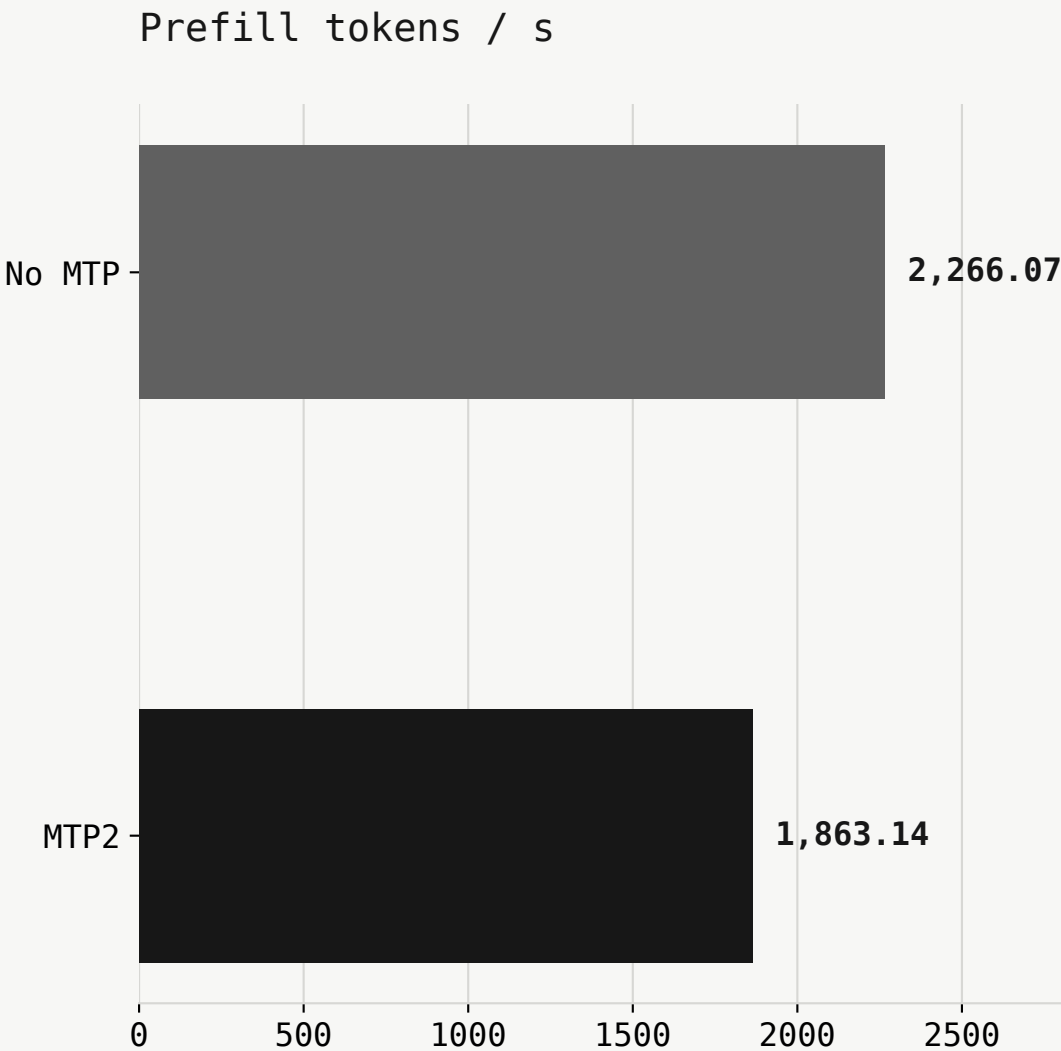
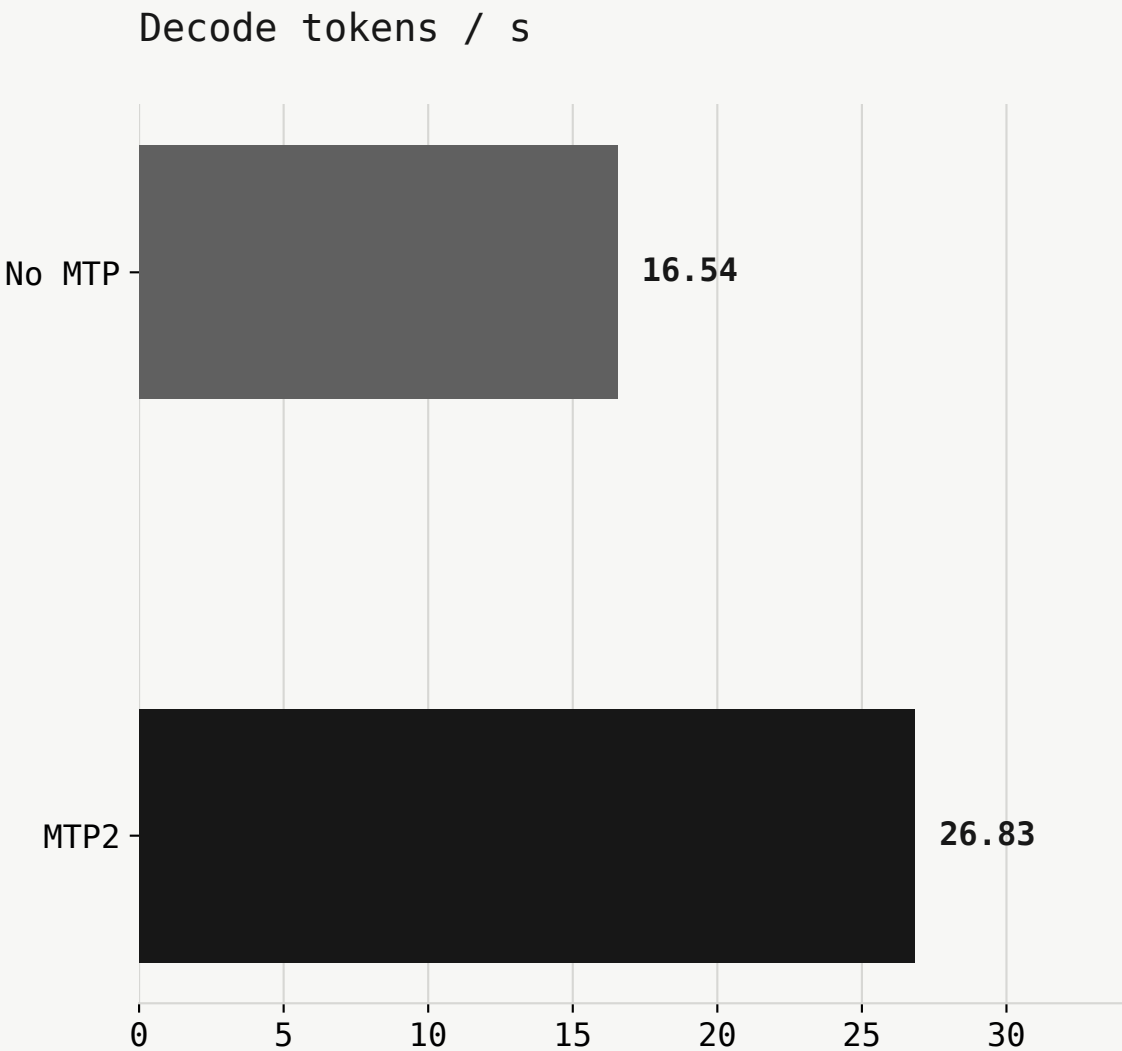
Qwen3.8 Flash Next: final DGX production

Verified 2026-09-05 18:19:19 MDT · original NVIDIA checkpoint · prefix caching ON

Engine	vLLM 0.28.1rc1.dev442+g7fbd44cbe.d20260905 base 7fbd44cbe0a90b9c8fd3a94a0f0401ac4b1bc719 + documented local patches
Runtime	PyTorch 2.13.0 · FlashInfer 0.6.18 · Transformers 5.16.1 Python 3.12 · driver 580.173.02 / CUDA 13.0 · kernel 6.17.0-1031-nvidia
Hardware	One GB10 · 20 ARM CPU threads · 121.6 GiB OS-visible unified memory TP1 · eager execution · original PLE FP8 mmap
Geometry	262,144 context · 131,072 output ceiling within remaining context 8 GiB BF16 KV · batch 2,048 · one sequence · MTP2
Architecture	125B main / 6B active + 51B n-gram embeddings + 4B MTP (NVIDIA card) 48 layers: 36 Gated DeltaNet + 12 sparse attention; 512 experts / 10 selected
Cache	Automatic prefix reuse ON · recurrent checkpoint grid 1,600 tokens align mode · retention interval 1,600 · separate QSA scratch ring 8
OpenCode	Exact served ID: nvidia/Qwen3.8-Flash-Next-NVFP4 · port 8092 Thinking medium; low/xhigh available · one image, video disabled

MTP2: the measured selection behind production

Historical cache-OFF comparison · 47,643 input · identical request and sampling · 131,072 configured context

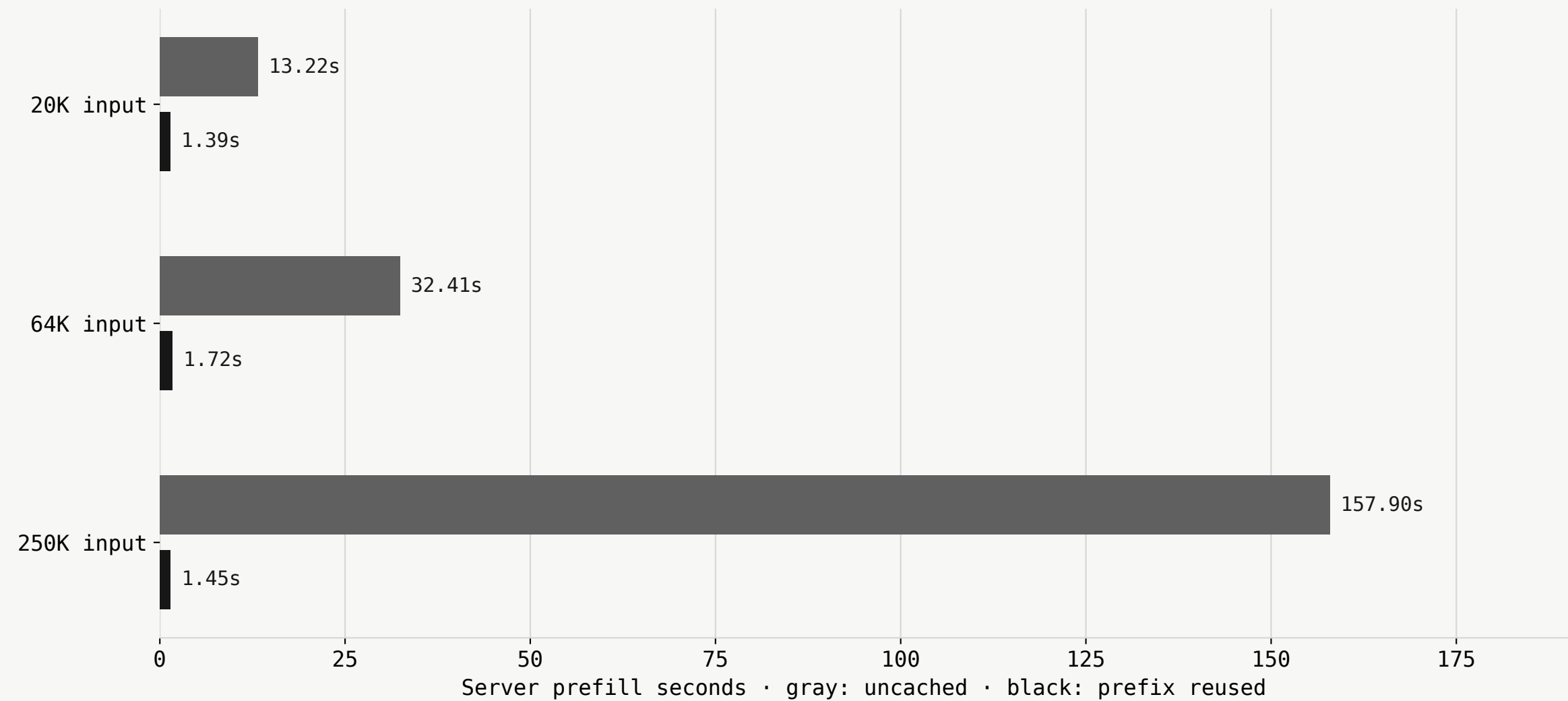


Decode +62.2% | Prefill -17.8%

One run per setting; includes reasoning. Historical selection evidence, not a new APC decode test.

Prefix cache: 250K prefill in about 1.5 seconds

Measured cold / warm pairs · thinking medium · exact answers verified · prefill timing, not decode speed

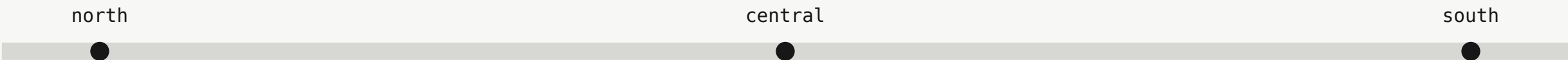


Reused tokens: 17,600 / 60,800 / 248,000

Cold controls use unique `cache_salt` namespaces. First cold request includes first-use effects.
Earlier-prefix edits and eviction can still cause misses; speedup applies to these repeated prompts.

250K needles: cold, warm, changed, and restored

Three audited keys embedded at roughly 6%, 50%, and 94% of the background records



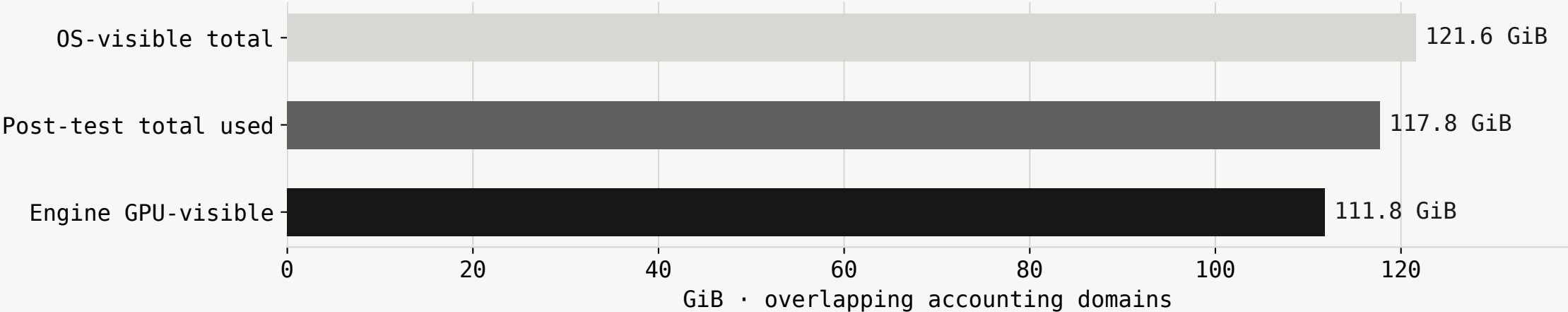
Condition	Input tokens	Cache hit tokens	Answer
cold	249,986	0	PASS
warm	249,986	248,000	PASS
suffix-warm	250,005	248,000	PASS
suffix-cold	250,005	0	PASS
original-after-suffix	249,986	248,000	PASS

Near-full boundary also passed: raw 260,834, managed 260,834 input tokens.

Suffix instruction asks for north/south only, then returns to all three.
This proves these retrieval cases; it is not exhaustive long-context reasoning or a multi-needle benchmark score.

Memory: prefix reuse inside the same 8 GiB pool

Unified-memory snapshots overlap; do not add GPU-visible and system-used measurements



KV allocation

8 GiB BF16 · 282,420 advertised cache tokens · max_num_seqs 1
1.08× capacity ratio is not tested concurrency

Checkpoints

Retention every 1,600 tokens stays inside the preallocated pool.
Denser retention changes cache eviction competition; cold requests still cost prefill.

PLE / pressure

47.684 GiB original FP8 PLE on disk; selected rows gathered through mmap.
Post-test swap 29.6%; memory headroom remains tight at full context.

Source fixes: what makes this deployment work

Original weights retained · official upstream sources for cache patches · open PRs are still local backports

PLE mmap / stride	Preserved eager-TP1 original-FP8 mmap reader. PR55375 handles non-contiguous state-index strides during speculative prefill/decode.
MTP ModelOpt	Remap draft-local layer0 to runtime layer48; preserve other mappings. Dispatch FP8_BLOCK_SCALES experts to block-FP8 MoE; main NVFP4 stays intact.
PR53798	Bind actual MambaSpec before the first request; restore recurrent state using its 1,600-token block size, not the generic attention block size.
PR54076	Materialize each crossed recurrent boundary. Local adaptation unwraps UniformTypeKVCacheSpecs for actual GDN and PLE groups; regression added.
PR55390 / 54713	Annotate the hybrid MTP group, then retain the lower checkpoint reachable after speculative lookup drops a block. Warning alone did not prove in-GPU misses.
Retention policy	--mamba-cache-mode align --prefix-cache-retention-interval 1600 Fixes the measured shorter-branch miss left by sparse retention0.
Client integration	Output131072; max_tokens null; thinking medium/low/xhigh. Preserve 1,200,000 ms general/header/chunk timeouts and provider options.
Validation	201 core/worker +186 cache/scheduler tests passed; source pre-commit passed. Two PP2 cases unavailable on one GPU. Earlier16 engine +2 wrapper tests retained.

Final production: complete launch command

Captured managed argv and allowlisted environment · all paths anonymized

ENVIRONMENT

```
#!/usr/bin/env bash
set -euo pipefail
cd /home/user/src/vllm-upstream
export PATH=/home/user/venvs/vllm/bin:/usr/local/cuda/bin:$PATH
export CUDA_HOME=/usr/local/cuda
export HF_HOME=/home/user/vllm/hf_cache
export HF_HUB_OFFLINE=1
export MAX_JOBS=1
export NVCC_THREADS=1
export TMPDIR=/home/user/vllm/tmp
export TOKENIZERS_PARALLELISM=false
export VLLM_ALLOW_LONG_MAX_MODEL_LEN=1
export VLLM_CACHE_ROOT=/home/user/vllm/cache
export VLLM_QWEN4_PLE_MMAP=1
export VLLM_USE_DEEP_GEMM=0
export XDG_CACHE_HOME=/home/user/vllm/xdg_cache
Existing production endpoint: port8092
```

Run the environment block first.
Do not start a second large engine.
Use the wrapper lifecycle for normal restarts.

Raw qualified command also included.
MAX_JOBS=1 and PLE mmap stay enabled.

SERVER

```
exec /home/user/venvs/vllm/bin/python -m vllm.entrypoints.openai.api_server \
  --model /home/user/models/nvidia-Qwen3.8-Flash-Next-NVFP4 \
  --served-model-name nvidia/Qwen3.8-Flash-Next-NVFP4 \
  --host 0.0.0.0 \
  --port 8092 \
  --dtype bfloat16 \
  --kv-cache-dtype auto \
  --max-model-len 262144 \
  --max-num-batched-tokens 2048 \
  --max-num-seqs 1 \
  --gpu-memory-utilization 0.8 \
  --quantization modelopt \
  --kv-cache-memory-bytes 8G \
  --speculative-config '{"method":"mtp","num_speculative_tokens":2}' \
  --reasoning-parser qwen3 \
  --enable-auto-tool-choice \
  --tool-call-parser qwen3_coder \
  --limit-mm-per-prompt '{"image":1,"video":0}' \
  --enable-prefix-caching \
  --mamba-cache-mode align \
  --prefix-cache-retention-interval 1600 \
  --skip-mm-profiling \
  --mm-processor-kwarg '{"max_pixels":1048576}' \
  --tensor-parallel-size 1 \
  --enforce-eager \
  --load-format safetensors \
  --safetensors-load-strategy lazy \
  --generation-config vllm \
  --no-enable-flashinfer-autotune \
  --override-generation-config '{"max_new_tokens":131072}'
```

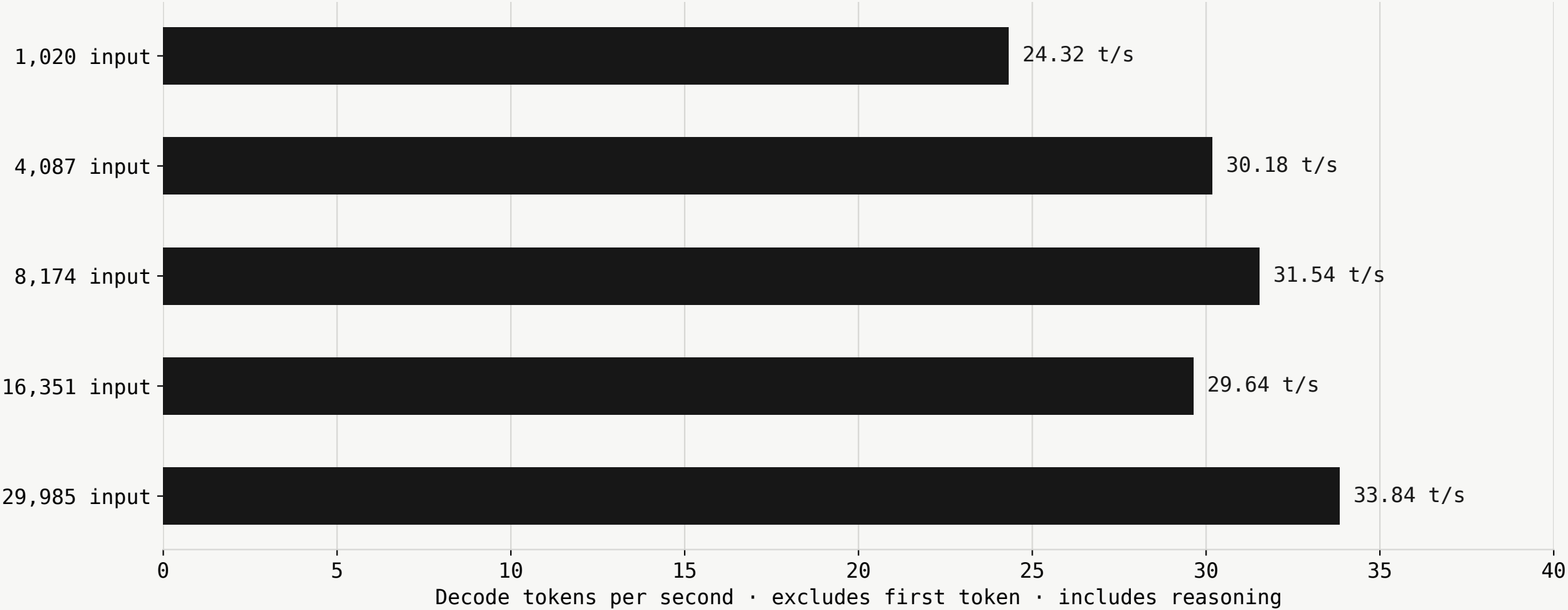
Validation: raw engine through actual OpenCode

Passed before final report generation · no inference submitted merely to redraw charts

Direct prefix suite	21 requests:20K/64K/250K cold, repeat, changed suffix, return; interleaved A/B/A, automatic inventory tool call and tool-result continuation.
State / boundary	Observed GPU state seeds match computed_tokens /1600 -1. Raw and managed260,834-token three-needle retrieval passed.
Managed engine	11 cache requests passed; actual argv matches wrapper command. Thinking, retrieval, tools, tool result, vision and prose passed.
TUI / OpenCode	Actual TUI warm action and streaming bash tool roundtrip passed. OpenCode aggregate cache hits: 3,200 tokens over 2 requests.
Production settings	262,144 context;131,072 output ceiling; BF16 KV; MTP2; APC ON. Thinking-only variants;20-minute transport timeouts; mirrors synchronized.
Limits	No extended soak, MTP3, concurrency>1, bitwise hidden-state proof, or131,072-token generated answer. Exact checks passed; broader claims are untested.

Decode through 30K: today's measured baseline

Historical pre-APC sweep · 262,144 configured context · MTP2 · BF16 KV · medium thinking

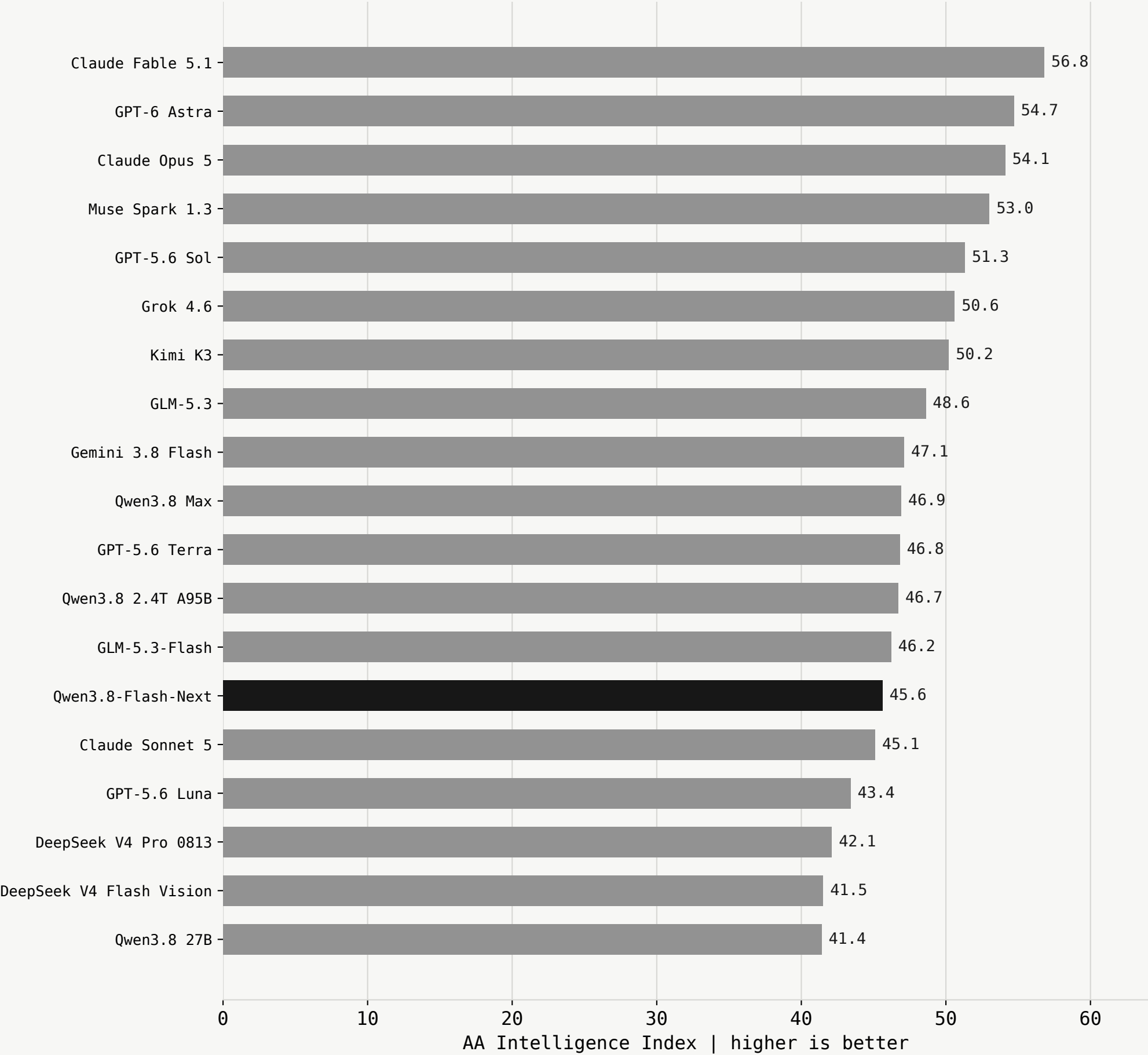


One warmed 256-token sample per length; ignore_eos was enabled.

These are retained cache-OFF measurements, not a fresh prefix-enabled speed sweep.
Fixed-length continuation can affect speculative acceptance; this is not an answer-quality benchmark.

Artificial Analysis: where the model sits

Intelligence Index v4.2 | leading model families | highest score only | one entry per model



Qwen3.8-Flash-Next: 45.6

Highest-scoring result per family across effort settings and older versions.
Selected effort and evaluation settings are recorded in the source data.